MILITARY SPECIFICATION

CLOTH, KNITTED, VINYON, AND VINYON-COTTON

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements for the procurement of vinyon and vinyon-cotton knitted cloth.

1.2 Classification - The cloth shall be furnished in the following types, as specified (see 6.2).

Type I - Vinyon knitted cloth

Type II - Vinyon-cotton knitted cloth

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-P-1134	Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (excluding Duck Fabrics)

STANDARDS

Federal

FED-STD-191	Textile Test Methods

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1491 Glossary of Knitting Imperfections

(When requesting specifications, standards, and publications, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids or request for proposal shall apply.

Rules and Regulations under the Textile Fiber Products Identification Act

(Applications for copies should be addressed to the Federal Trade Commission, Washington, D. C. 20580.)

American Society for Testing and Materials

ASTM D276 Identification of Fibers in Textiles

ASTM D1577 Test for Linear Density of Textile Fibers

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Materials

3.1.1 Vinyon fiber- The fiber shall be vinyon, 2-1/2 denier having a staple length of not less than 1-1/2 inches. The infrared spectra of vinyon shall be as shown in Figure 1.

3.1.2 Cotton - The cotton used in the manufacture of the yarn shall be of suitable length and grade to meet the requirements of the specification.

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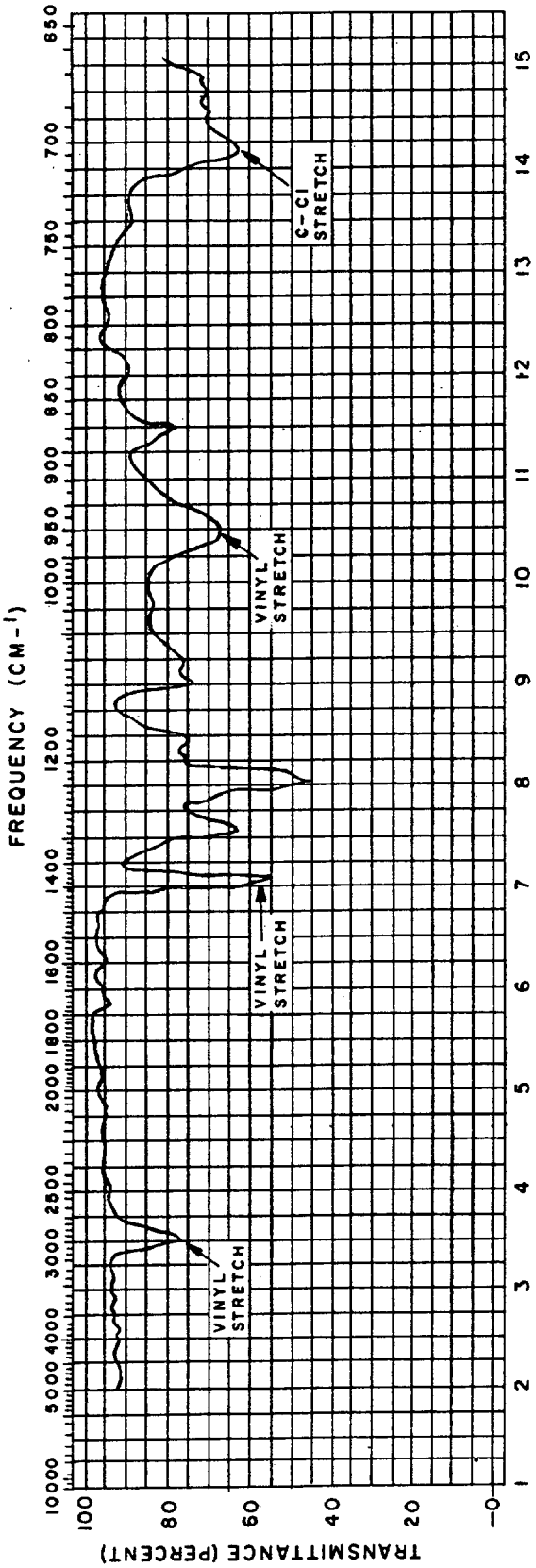


Figure 1. Infrared spectra of vinyon fiber.

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3.2 Yarn

3.2.1 Type I - The yarn shall be evenly spun from vinyon fibers into single yarn, size 36 ± 5 (tex units).

3.2.2 Type II - The yarn shall be spun from a fiber blend of not less than 35 percent cotton and 60 plus or minus 5 percent vinyon, size 25 ± 5 (tex units).

3.3 Color - Unless otherwise specified, the color of the finished cloth shall be as specified by the procuring activity.

3.4 Physical requirements - The finished cloth shall conform to the requirements specified in Table I.

TABLE I

PHYSICAL PROPERTIES

CHARACTERISTICS	REQUIREMENTS	
	Type I	Type II
Type of Knit	1 by 1 rib circular knit	1 by 1 rib circular knit
Texture per inch, min.		
Wales	20	20
Courses	30	30
Bursting strength, pounds, min.	50	--
Shrinkage after five launderings, percent, max.		
Wales	5.0	--
Courses	5.0	--

3.5 Width - The width of the finished cloth shall be specified by the procuring activity (see 6.2).

3.6 Length and put-up - The cloth shall be furnished in continuous lengths, each not less than 35 yards. Each length shall be put-up on full width rolls as specified in PPP-P-1134.

3.7 Fiber identification - Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

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3.8 Workmanship - The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance - Where certificates of compliance are submitted in accordance with 4.3, they shall contain verifiable actual test and inspection data. The Government reserves the right to inspect and test the cloth to verify the validity of the certification.

4.2 Inspection - Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection - In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, notified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item - Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.2.

4.2.2.1 Yard-by-yard examination - The required yardage of each roll shall be examined and defects clearly noticeable at normal inspection distance (3 feet) shall be classified as listed in Table II and as stated in MIL-STD-1491. All defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only one of these shall be counted. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be one linear yard. The acceptable quality level (AQL) shall be 6.5 defects per 100 units (yards). The sample size shall be in accordance with Inspection Level II of MIL-STD-105. The lot size shall be expressed in units of one yard each. The number of rolls

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selected shall be in accordance with Table III. An approximately equal number of yards shall be examined in each roll in the sample.

TABLE II

DEFECTS

Hole, cut, tear, smash
Mend or snag
Missing yarn or miss-knit
Abrasion mark resulting in weak area
Heavy or thin yarn, more or less than twice the normal thickness
Slub, slug, or kink
Embedded crease
Any knot extending above the face of the fabric
Knitting tension (stretch) too tight or too loose
Spot or stain (any)
Barre or tiger stripe

4.2.2.2 Overall examination - Each defect shall be counted not more than once in each roll examined. The sample units shall be one roll. The sample size (number of rolls selected as sample) and the number of defects acceptable shall be shown in Table III.

OVERALL DEFECTS

Off shade
Width less than specified
Cut edge is not parallel to wales in finished cloth
Overall uncleanness
Uneven knitting throughout the roll
Not labeled in accordance with the Textile Fiber P cs Identification Act

TABLE III

SAMPLE SIZE

	ple Size (Rolls)	Maximum Number of Defects Acceptable in Sample
Up to 1200 1/	3	0
1,201 up to and including)	5	0
3,201 up to and includi ,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,000 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

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4.2.2.3 Examination for length

4.2.2.3.1 Individual rolls - During the yard-by-yard examination each roll shall be examined for gross length. Any gross length found to be less than the minimum specified or more than 2 yards less than the gross length marked on the ticket shall be considered a defect with respect to length. The sample unit shall be one roll. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length.

4.2.2.3.2 Total yardage in sample - The lot shall be unacceptable if the total of the actual gross length of rolls in the sample is less than the total of the gross lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.2.2.3.3 Examination for compliance with Textile Fiber Products Identification Act - During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.2.3 Examination of preparation for delivery requirements - An examination shall be made in accordance with provisions of PPP-P-1134 to determine that packaging, packing, and marking requirements of Section 5 of this specification are complied with.

4.3 Testing of the end item - The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table IV and the following subparagraphs shall be followed. The physical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 3-1/2 yards full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual value utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

Lot size, yards	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE IV

TEST METHODS

Characteristic	Req. Para.	Test Method
Material Identification:		
Vinyon	3.1.1	1/ 2/
Cotton	3.1.2	1200 1/
Denier	3.1.1	4021 1/
Tex units	3.2.1 and 3.2.2	4021 1/
Fiber content	3.2.2	1/
Texture (wales and courses)	3.4	5070
Bursting strength	3.4	5120
Type of knit	3.4	Visual
Shrinkage (5 complete launderings)	3.4	5554 3/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ ASTM D276. Identification of Fibers in Textiles-Infrared Spectra of man-made Fibers (Dissolve the sample in 1,2 dichloroethane. Filter the solution through No. 41 Whatman paper and a regular funnel. Cast the fiber solution on fused sodium chloride plate and place in spectrophotometer for analysis).

3/ except that -

a. the water temperature shall be 46 ± 1°C.

b. the detergent shall meet the requirements of MIL-D-43362, Detergent, Laundry (Anionic - A Standard For Testing.)

c. the wash cycle shall be 30 minutes. The specimen shall then be extracted in a tunble dryer at a maximum temperature of 49°C.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Levels A and C - The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

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5.2 Packing - Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C - The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use - The cloth is intended for use as basic material for flyers' drawers and undershirt with the ventilated wet suit.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title and date of this specification
(b) Type and color required (see 1.2 and 3.3)
(c) Width required (see 3.5)
(d) Length required (see 3.6)
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

6.3 Supersession data - This specification includes the requirements of AS 2608 of 3 June 1970.

Preparing Activity
Navy - AS

Project No. 8305-N085

☆ U. S. GOVERNMENT PRINTING OFFICE: 1973 -713-144/126

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DOCUMENT IDENTIFIER AND TITLE CLOTH, KNITTED, VINYON, AND VINYON-COTTON
MIL-C- 81938 (AS)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

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